

Individual Assignment 5

Zelda Reif

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1. Set Up

Packages

```
# general use
library(tidyverse)
library(here)
library(janitor)
library(snakecase)

# wrap axis labels
library(scales)

# reading .xlsx files
library(readxl)

# visualize missing data
library(naniar)

# arrange plots
library(patchwork)
```

Data

```
# taxonomic info
taxon_list <- read_csv(here("data","taxon_list.csv"))
```

```

# aquatic invertebrates
aq_ins <- read_xlsx(
  here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "Aquatic Insects"
)

# site info
sites <- read_xlsx(
  here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "sites"
)

# water quality
water_quality <- read_xlsx(
  here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "Water Quality",
  na = c("", "n/a", "N/A", "over", "173+")
)

```

2. Cleaning

sites object

```

# creating new object
ncos_sites <- sites |>
  # filtering to only include ncos quarterly sampling sites
  filter(sector == "NCOS" & sampling_frequency == "Quarterly")

```

clean taxon_list

```

taxon_list_clean <- taxon_list |>
  # convert taxon names to snakecase
  mutate(verbatim_name = to_any_case(verbatim_name,
    case = "snake"))

```

clean water_quality

```
water_quality_clean <- water_quality |>
  clean_names() |>
  semi_join(ncos_sites, by = "site") |>
  unite("date_site", date, site, remove = FALSE) |>
  group_by(date, site, date_site) |>
  summarize(
    med_ph = median(p_h, na.rm = TRUE),
    med_sal = median(salinity_ppt, na.rm = TRUE),
    med_do = median(dissolved_oxygen_mg_l, na.rm = TRUE)) |>
  ungroup() |>
  filter(date_site != "2022-08-12_NVBR")
```

clean aq_ins

```
# creating new clean object from aquatic inverts
aq_ins_clean <- aq_ins |>
  # clean column names
  clean_names() |>
  # filtering join: only include sites occurring in ncos_sites
  semi_join(ncos_sites, by = c("site")) |>
  # renaming columns
  rename(date = date_on_vial) |>
  # select columns of interest
  select(
    date, sample_type, site,
    ostracod,
    copepod,
    hemiptera_corixidae_boatman,
    cladocera,
    nematode,
    diptera_ceratopogonidae,
    annelida_oligochaete,
    diptera_chironomid,
    annelida_polychaete) |>
  # filter to only include "filtered beaker" samples
  filter(sample_type %in% c("FB 250", "FB250")) |>
  # convert the data frame to long format
  pivot_longer(
    cols = ostracod:annelida_polychaete,
    names_to = "taxon",
    values_to = "abundance") |>
```

```

# group by date, site, taxon
group_by(date, site, taxon) |>
# calculate average abundance per liter (sum abundance divided by 7.5
  ↳ liters)
summarize(ave_lit = sum(abundance, na.rm = TRUE)/7.5) |>
# ungroup data frame
ungroup() |>
# create a new column called `date_site` to join with water quality
unite("date_site", date, site, remove = FALSE) |>
# join with water quality
left_join(
  select(water_quality_clean,
        date_site,
        med_ph,
        med_sal,
        med_do),
  by = "date_site"
) |>
# join with taxon list
left_join(taxon_list_clean, by = c("taxon" = "verbatim_name")) |>
# add full names of sites
mutate(site_full = case_when(
  site == "NVBR" ~ "North of Venoco Bridge",
  site == "NEC" ~ "South of Venoco Bridge",
  site == "NMC" ~ "Main Channel",
  site == "NPB" ~ "South of Phelps Bridge",
  site == "NPB1" ~ "North of Phelps Bridge",
  site == "NPB2" ~ "Phelps Road",
  site == "NWP" ~ "West Pond",
  site == "NDC" ~ "Devereux Creek"
)) |>
# setting factor levels for site
mutate(site_full = fct_reorder(site_full, med_sal, .fun = "median", .na_rm
  ↳ = TRUE),
  site_full = fct_rev(site_full))

```

Problems

1. Visualizing differences across sites in major taxon abundance

a. Plan your figure

- x-axis: site
- y-axis: ostracod abundance
- geometry to represent average abundance/L: `geom_jitter`
- geometry to represent medians: `stat_summary()`

b. Wrangle your data

```
# create new object from aq_ins_clean
ost_abundance <- aq_ins_clean |>
  # filter to only ostracods
  filter(taxon == "ostracod") |>
  # remove missing abundance/L values
  filter(!is.na(ave_lit)) |>
  # group by site
  group_by(site_full) |>
  # log + 1 transform mean abundance per liter
  mutate(log_ave_lit = log(ave_lit + 1)) |>
  # ungroup dataframe
  ungroup()

slice_sample(ost_abundance, n = 10)
```

A tibble: 10 x 24

	date_site	date	site	taxon	ave_lit	med_ph	med_sal	med_do
	<chr>	<dtm>	<chr>	<chr>	<dbl>	<dbl>	<dbl>	<dbl>
1	2022-02-23_NPB1	2022-02-23 00:00:00	NPB1	ostr~	0	NA	NA	NA
2	2023-05-05_NPB	2023-05-05 00:00:00	NPB	ostr~	24.7	NA	NA	12.1
3	2021-11-12_NPB	2021-11-12 00:00:00	NPB	ostr~	79.2	NA	NA	NA
4	2022-05-06_NMC	2022-05-06 00:00:00	NMC	ostr~	0	9.96	1.45	16.5
5	2022-08-08_NPB2	2022-08-08 00:00:00	NPB2	ostr~	2	7.68	2.36	2.43
6	2023-11-14_NPB	2023-11-14 00:00:00	NPB	ostr~	18	NA	NA	NA
7	2024-01-28_NPB	2024-01-28 00:00:00	NPB	ostr~	3.33	7.43	3.08	6.6
8	2022-01-21_NEC	2022-01-21 00:00:00	NEC	ostr~	0	NA	NA	NA
9	2022-01-28_NDC	2022-01-28 00:00:00	NDC	ostr~	9.47	7.65	2.00	9.55

```

10 2023-07-17_NPB  2023-07-17 00:00:00 NPB  ostr~    4.53  NA      NA      NA
# i 16 more variables: taxonRank <chr>, scientificName <chr>, kingdom <chr>,
#   Phylum <chr>, Class <chr>, Subclass <chr>, Superorder <chr>, Order <chr>,
#   Infraorder <chr>, Family <chr>, Subfamily <chr>, Tribe <chr>, Genus <chr>,
#   comments <chr>, site_full <fct>, log_ave_lit <dbl>

```

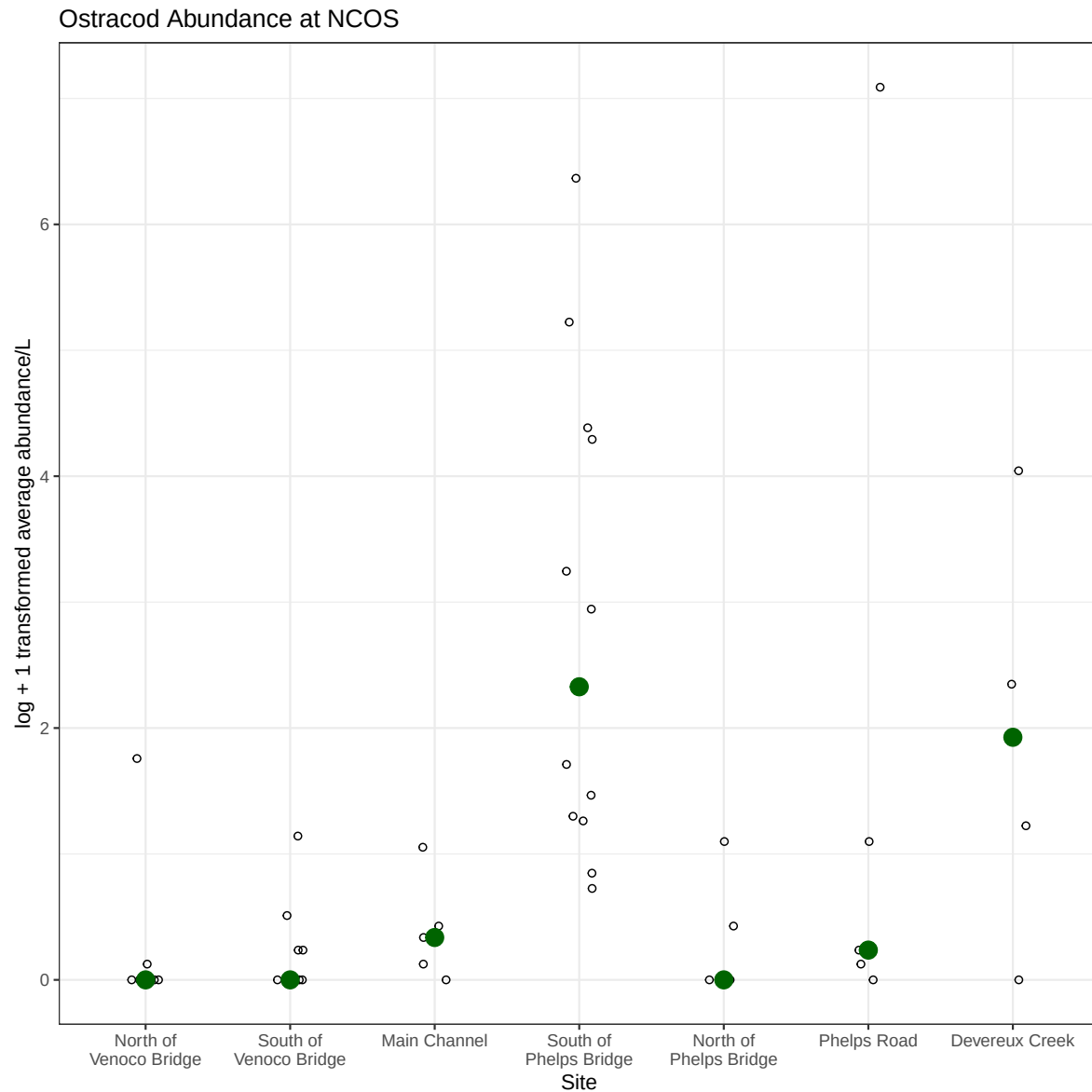
c. Code your figure

```

# base layer
ost_abundance <- ggplot(data = ost_abundance,
                        # x-axis: site
                        # y-axis: mean abundance/L
                        mapping = aes(x = site_full,
                                      y = log_ave_lit)) +
# first layer: points to represent observations
geom_jitter(height = 0,
            shape = 21,
            width = 0.1) +
# second layer: points to represent medians
stat_summary(geom = "point",
            fun = "median",
            size = 4,
            color = "darkgreen") +
# wrapping x-axis text
scale_x_discrete(labels = label_wrap(15)) +
# relabelling x- and y-axis, adding title
labs(x = "Site",
     y = "log + 1 transformed average abundance/L",
     title = "Ostracod Abundance at NCOS") +
# different theme
theme_bw()

# displaying object
ost_abundance

```



d. Create a multipanel figure

e. Interpret your figure

differences between sites in copepod and ostracod abundance how ostracod abundance may relate (or not) to site-level differences in salinity

2. Visualizing total abundance as a function of salinity

a. Plan your figure

x-axis: y-axis: color: geometry: facets:

b. Wrangle your data

c. Code your figure

d. Interpret your figure

In 3-5 sentences, describe differences or similarities between taxa in relationships between abundance and salinity. Be specific using the visual components of the plot.